

Jet & Airflow Dyeing Machine Hydraulic & Mechanical Constraints

1. Overview of Hydraulic & Aerodynamic Transport Constraints

In rope dyeing machinery (Jet, Soft-Flow, and Air-Jet), fabric movement is driven by liquor pressure (hydraulic) or compressed air streams (aerodynamic) combined with a motorized winch reel. The primary engineering goal is to maintain uniform dye migration while preventing fabric surface abrasion, pilling, rope creasing, and mechanical tangling.

2. Mathematical Formulas & Parameters

Rope Length: $L = (W_{\text{fabric}} * 1000) / (GSM * W_{\text{width}})$
Turn Time: $t_{\text{turn}} = L / v_{\text{fabric}}$. *Target: 0.8 to 1.5 minutes (48-90 seconds).*

3. Nozzle & Hydraulic Constraints

Substrate / GSM	Nozzle Dia (mm)	Nozzle Pressure (bar)	Reel Speed (m/min)	Primary Machine Risk
Light Knits (100-180 GSM)	90 - 110 mm	0.2 - 0.5 bar	200 - 300 m/min	Pilling, stretch, filament breakage
Medium Knits (180-300 GSM)	70 - 90 mm	0.6 - 0.9 bar	300 - 400 m/min	Rope marks if turn time > 90s
Heavy Woven (300-500+ GSM)	50 - 70 mm	1.0 - 1.4 bar	250 - 350 m/min	Chafing, high nozzle impulsion

4. Machine Safeguards & Alarms

Alarm Sensor	Trigger Condition	Automated Action	Remediation
Seam Detector (Magnet)	Turn time > 120 s	Boosts reel speed	Reduce loading weight
Knot / Tangle Sensor	Mechanical rope stop	Main pump & reel stop	Prevents fabric tearing
Pressure Differential Sensor	Filter delta P > 0.6 bar	Auditory alarm	Clean liquor filter mesh